

Forest Fire Weather Index (FWI) Prediction Project

1. Problem Statement

This project predicts the Fire Weather Index (FWI) using weather conditions and forest fuel information. The objective is to estimate the potential danger of a forest fire before it becomes severe. Early prediction helps forest authorities take preventive measures, allocate resources, and reduce environmental and human losses.

2. Input Features and Full Forms

Feature	Full Form	Short Explanation
Temperature	Temperature	Air temperature in °C.
RH	Relative Humidity	Percentage of moisture in the air.
Ws	Wind Speed	Wind speed; higher wind can spread fire faster.
Rain	Rainfall	Amount of rainfall in mm.
FFMC	Fine Fuel Moisture Code	Index showing how dry fine surface fuels (leaves, grass) are.
DMC	Duff Moisture Code	Index showing the moisture of the upper organic soil layer.
DC	Drought Code	Index representing long-term dryness of deep soil and large fuels.
ISI	Initial Spread Index	Index estimating how quickly a fire can spread.
BUI	Build Up Index	Index representing the total amount of available fuel.
Region	Region	Forest area (Bejaia or Sidi Bel-Abbes).
Classes	Fire Class	Whether the observed condition is 'fire' or 'not fire'.

3. What is FWI?

FWI (Fire Weather Index) is a numerical index that estimates the potential intensity and danger of a forest fire. Higher FWI values indicate greater fire risk and faster fire spread.

4. FWI Danger Levels

FWI Range	Danger Level
0 - 5	Very Low
5 - 15	Low to Moderate
15 - 30	High
30 - 45	Very High
45 and above	Extreme